

Judge Agent / Evidence Review Engine

Compact technical design summary for the Gemma hackathon project

Core principle: The Judge compares evidence, not opinions, and produces the most balanced conclusion justified by the available sources.

1. Final output returned to the user

The Judge generates one structured answer. Each field corresponds to a question a real user naturally has:

Output field	Meaning
direct_answer	One concise and balanced answer.
potential_benefits	Benefits supported by the available evidence.
potential_risks	Risks or disadvantages supported by the evidence.
who_may_benefit	Populations for whom benefits appear more likely.
who_should_be_cautious	Groups for whom caution is justified.
conditions_that_change_the_answer	Context such as age, health status, dosage, protocol, food quality, or lifestyle.
evidence_strength	strong / moderate / limited / conflicting / insufficient
remaining_uncertainties	Important gaps or unresolved conflicts.
why_this_conclusion	Short, user-facing explanation of the evidence basis.
practical_conclusion	Useful real-world interpretation without unsafe personalized medical advice.

2. Example: Intermittent fasting

Direct answer	Intermittent fasting may be helpful for some adults, but it is not universally beneficial or clearly superior to other sustainable dietary approaches.
Potential benefits	May support modest weight loss; may improve some metabolic markers; may provide a simple eating structure.
Potential risks	Hunger, fatigue, difficulty concentrating, poor adherence, or overeating during the eating window.
Who may benefit	Some healthy adults and some adults with overweight who prefer structured eating schedules.
Who should be cautious	People with eating-disorder history, pregnancy or breastfeeding, children, and people using glucose-lowering medication.
Conditions that change the answer	Fasting protocol, total calories, food quality, medical history, medication use, and long-term sustainability.
Evidence strength	Moderate.
Remaining uncertainties	Long-term outcomes and whether one fasting protocol is clearly better than another.
Why this conclusion	Several controlled human studies show modest benefits, but benefits are not consistently greater than conventional calorie restriction, and long-term evidence remains limited.
Practical conclusion	It can be one reasonable dietary strategy, but not the only effective option.

3. What the Judge receives from the agents

Each agent provides three things:

- An overall conclusion from its own perspective.
- Atomic claims, each linked to one or more source IDs.
- Evidence sources with metadata and excerpts.

The generic source model supports research papers and wider web data, including:

- systematic reviews and trials
- clinical guidelines and government pages
- medical organizations and academic sources
- datasets and statistics reports
- expert and news articles
- company pages
- blogs, forums, and social media

4. Evaluation criteria

Criterion	How it is used
Evidence quality	How reliable is the source type and methodology?
Question relevance	Does the evidence answer the exact user question?
Population match	Does the studied or described population match the user context?
Methodological strength	Are the design, sample, duration, comparison and outcome measures adequate?
Claim support	Does the cited source actually justify the claim?
Consistency	Do independent sources reach similar conclusions?
Independence	Are sources truly independent or all repeating the same original evidence?
Recency	Is the information current enough for the topic?
Limitations	Are important weaknesses acknowledged?
Conflict of interest	Could commercial or institutional incentives affect the claim?
Benefits versus risks	What positive and negative effects are genuinely supported?
Remaining uncertainty	What cannot be concluded confidently?

5. How evidence strength is determined

The Judge does not count sources mechanically. A large number of weak or duplicated pages must not outweigh a smaller amount of strong, independent evidence.

Strong: Several high-quality, relevant, independent sources are consistent, with limited major weaknesses.

Moderate: The general direction is supported, but limitations, population gaps, or some inconsistency remain.

Limited: Some support exists, but evidence is weak, indirect, small, short-term, or based mainly on lower-quality sources.

Conflicting: Credible sources disagree and the conflict cannot be resolved confidently.

Insufficient: There is too little reliable and relevant evidence to form a dependable conclusion.

6. Source-specific use

Research papers	Used for causal, clinical, and quantitative claims. Judge reviews design, sample, duration, population, outcomes, and limitations.
Guidelines / government / medical organizations	Used for current recommendations and safety guidance. Judge checks issuing body, review process,

	target population, references, and update date.
Datasets / statistics	Used for prevalence, trends, and measured outcomes. Judge checks origin, collection method, coverage, and interpretation.
News / expert articles	Useful for explanation and recent developments, but should link to original evidence and avoid exaggerated headlines.
Company sources	Useful for product details or internal data, but commercial incentives and independent validation matter.
Blogs / forums / social media	Useful for experiences, practical barriers, and reported concerns; not sufficient alone for strong medical or causal conclusions.

7. Reasoning safeguards

- Detect cherry-picking or ignored conflicting evidence.
- Detect correlation presented as causation.
- Detect overstated or unsupported conclusions.
- Detect population overgeneralization.
- Detect animal evidence generalized directly to humans.
- Detect anecdotal evidence treated as prevalence data.
- Detect articles repeating the same original source.
- Detect circular citation chains and company-only support.
- Prefer quality and independence over the number of sources.
- Allow conflicting or insufficient evidence as legitimate outcomes.

8. Processing workflow

1. Receive user question and structured agent outputs.
2. Validate required fields and preserve unknown values.
3. Normalize URLs, titles, dates, and source types.
4. Deduplicate by DOI, canonical URL, normalized title, or original-source relationship.
5. Evaluate each claim against its cited sources.
6. Compare evidence quality, relevance, population, consistency, and independence.
7. Identify supported benefits, risks, population differences, conditions, and uncertainties.
8. Assign overall evidence strength.
9. Generate and validate the final structured JSON answer.
10. Retry once for malformed JSON; use a safe fallback if generation fails.

9. Simple technical split

Deterministic Python	Gemma semantic reasoning
Schema validation	Claim-to-source support
Missing-field handling	Relevance and population interpretation
URL and date normalization	Overstatement detection
Deduplication and source-chain detection	Conflicting-evidence comparison
Basic source-type rules	Benefit/risk synthesis
Output parsing and validation	Final answer generation

Result: one balanced answer that clearly communicates benefits, risks, who it applies to, and how certain the evidence is.